PowerPack Engine Troubleshooting Guide

Tags	web-ui-12.1.2-484.el8.x86_64.rpm, master.dynamic_app_index_map, updateDeviceClass, APIError, master.definitions_dev_classes
Team Owner	Sprint Eastwood
Status	ACTIVE
Slack Channel	#help-powerpack-engine

About This Guide

This guide is intended to help diagnose issues with the PowerPack engine.

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Issues related to (Import/Install/Export Issues)

These issues usually take the form of being unable to import/install/export a PowerPack, or unintended behavior arising immediately after doing so.

Before you escalate an issue, capture and include the PHP Developer logs in your ticket. Steps on how to obtain the logs are provided in the "Capturing the Useful Logs for An Issue" section. Some of the common issues listed in this guide include the log error that appeared when they were solved before.

Capturing the Useful Logs for an Issue

When reporting an issue regarding the PowerPack engine to Engineering, be sure to include the PHP Developer logs. These logs contain useful information that assists in getting escalations resolved quickly. Follow the below steps to gather the logs. They will show you how to enable PHP Developer logs on the appliance, and capture `em7php.log`, `php-error.log`, and `mysqli.log`.

1. Go to System → Tools → PHP Developer Logs
2. Under "UI Developer Log Levels", make sure the "Debug" box is checked.
3. At the bottom of the page, set how long you want log data to be captured. (Next to "Set Log Config").
4. Make sure `em7php.log`, `mysqli.log`, and `php-error.log` have the correct permissions (664: `-rw-rw-r--`).
5. Truncate the old `em7php.log`, `mysqli.log`, and `php-error.log`.
6. Reproduce the issue.
7. Download all three newly generated logs.

> If you search for the term `error` in em7php.log, you can get the exact log message that appeared when the the import/export failed.

Previously Reported Issues

Content in the PowerPack is Partially Updated on Import

If you notice that some content from the PowerPack is only partially updated when a new version of the PowerPack is imported/installed, the `Enable Selective PowerPack Field Protection` may be enabled.

1. Go to System → Settings → Behavior
2. Check whether the `Enable Selective PowerPack Field Protection` is enabled. This setting prevents some fields from being updated while installing a PowerPack.
3. You can check which fields are included in this list by running the following query: `(SELECT * FROM master.definitions_ppack_config_fields;)`

Starting from version 11.2.0, the "Enable Selective PowerPack Field Protection" is also applied on a per PowerPack basis. There is no option in the UI to control which fields should be updated when "Enable Selective PowerPack Field Protection" is enabled. This can be done by updating the database table field using SQL.

> Refer to EM-49181: Event policy only partially updated after import new version of the powerpack CLOSED for more information.

504 Gateway Timeout Error

Sometimes the UI gives a 504 error when installing a PowerPack, but the backend process continues to install the PowerPack. You can confirm whether or not the PowerPack was successfully installed by checking the logs tab of the default organization.

1. Go to Registry → Account → Organization.
2. Click on the wrench icon of the organization with id 0 (default).
3. Go to the logs tab.
4. In the search bar, select `Search Message`, type keyword "PowerPack", and click "Search".
5. The list of PowerPack uploads and install logs will be shown.
6. If your PowerPack is installed, its name will show up here.

If the PowerPack is not installed:

The PowerPack could be large and take too much time to process, causing a timeout while the install is still in progress. Increasing `fastcgi_read_timeout` in the `nginx` configuration file can circumvent this error.

File Location: `/opt/em7/share/config/nginx.d/em7ngx_web.fragment`

Modifying the configuration file:

1. Locate the file at `/opt/em7/share/config/nginx.d/em7ngx_web.fragment`
2. Create a backup of this file: `cp /opt/em7/share/config/nginx.d/em7ngx_web_ui.fragment /opt/em7/share/config/nginx.d/em7ngx_web_ui.fragment.bak`
3. Increase `fastcgi_read_timeout`. This value is the time in seconds before a timeout will occur.
4. Once updated you will need to restart the `php-fpm` service using `cmd - systemctl restart php-fpm`

```
location ~ \.em7 {
  fastcgi_split_path_info ^(.+\.em7)(/.+)$;
  fastcgi_pass unix:/var/run/php-fpm/em7.sock;
  fastcgi_index index.em7;
  fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/ap/www$fastcgi_script_name;
  fastcgi_param SSL_CLIENT_VERIFY $ssl_client_verify;
  fastcgi_param SSL_CLIENT_S_DN $ssl_client_s_dn;
  fastcgi_param SSL_CLIENT_S_DN_CN $ssl_client_username;
  fastcgi_param SSL_CLIENT_CERT $ssl_client_raw_cert;
  fastcgi_read_timeout 360;
  include fastcgi_params;
}
```

Refer to EM-46138: Powerpack installation is returning with 504 CLOSED for more information.

PowerPack Contains Errors and Cannot Be Compiled

Possible Log Error: (em7_php.log) "Failure to initialize item of type [alert] w/identifier[XXXX]"

This error can occur when an event policy references an alert that is not present on the system. The PowerPack engine tries to fetch this alert in `master.dynamic_app_alerts`, but that specific alert ID is not present. This results in unsuccessful PowerPack compilation.

The solution is to delete the offending event policy and corresponding alerts, then recreate them.

The following SQL query can help you determine which event policy the alert is linked to (Replace EVENT_ID with an event policy ID):

```
SELECT * FROM master.policies_events WHERE id = 'EVENT_ID';
```

In this query result, `Xid` is the app id and `Xoid` is the alert id.

Refer to EM-45661: PowerPack contains errors and cannot be compiled CLOSED and EM-48783: Powerpack cannot add Runbook Policies CLOSED for more information.

Unable to Build the PowerPack: Server Error

Possible Log Error: (php-error.log) PHP Fatal error: Allowed memory size of 536870912 bytes exhausted (tried to allocate 62914560 bytes) in /opt/em7/phpshare/classes/db/mysqli_helper.inc on line 715

The workaround for these types of issues is to increase the memory limit of the script. This has been fixed in 11.2.0 (EM-21603) and 11.1.1 (EM-47827). For prior versions, you can increase the memory limit by setting the `ppack_memory_limit` value in `/etc/silo.conf`. This value sets the memory runtime during PowerPack import/install. Below are the steps to follow:

1. Open `/etc/silo.conf` to edit it.
2. Add the `ppack_memory_limit` in silo.conf in the [LOCAL] section, as shown below. This change will set the memory limit as per the configuration in the above file.
3. Try to rebuild the PowerPack.
4. Once the build is complete, make sure to update `memory_limit` in php.ini to its original value.

```
[LOCAL]
rootdir = /opt/em7
vardir = /var/lib/em7
logdir = /var/log/em7
rundir = /run/em7
ipaddress = 10.2.15.70
dbdir = /data/db
dbserver = 127.0.0.1
dbport = 7706
dbuser = clientdbuser
dbpasswd = "nETs1fMYao8="
portcheck = /usr/bin/nmap
model_type = 1
eventmanager = internal,api,email,syslog,trap,dynamic,agent
dbsecureiv = "hbNBZn0POuMWMUkVJbzxAg=="
kbase_disabled = 1
ppack_tree = /data/ppdev
ppack_memory_limit=1G

[SLADMIN]
sladmin_user = em7admin
sladmin_cred = Arl_RIU6XD2h0KA7pBiJB13foXJZRCOloTyOfu6wgL8z32vUf7L4Xe5Ehqz3ZTfn
```

Adding `ppack_memory_limit` to silo.conf

Refer to : EM-53444: Unable to build PP revision CLOSED for more information.

"Could Not Unpack PowerPack File for Installation": Duplicate Table Entries

Possible Log Error: (em7php.log) |error| could not unpack source with provided encryption key

This issue can arise when the table `master_filestore.metadata_powerpack` contains two records with the same `ppguid` and `revision`. One of these records has old digest data, so when the PowerPack is imported, it fetches the old record and updates it on the installer. There is a difference between the PowerPack being imported and the value fetched from the database, which means the PowerPack is not allowed to be installed. These database table entries do not get cleared even when the PowerPack is deleted.

First, check to see if there are two PowerPack entries in `master_filestore.metadata_powerpack` using the following SQL query (Replace PP_GUID with the GUID of the PowerPack). If this is the case, continue with the steps below. If not, this section is not relevant to your issue.

```
SELECT * FROM master_filestore.metadata_powerpack WHERE ppguid LIKE 'PP_GUID'
```

The following steps are a workaround to install the PowerPack. The duplicate row from the table `master_filestore.metadata_powerpack` must be deleted. In the queries below, replace FILE_ID with the file ID of the duplicate table entry and PP_GUID with the GUID of the PowerPack.

Make sure to properly backup `master_filestore.metadata_powerpack` as described below. If you accidentally delete the wrong table entry, it will help you restore the table easily.

1. Create a backup of the `master_filestore.metadata_powerpack` table, using the command below. Use the terminal in CDB.

```
$>mysqldump --user=<username> --password=<password> --lock-tables master_filestore metadata_powerpack >
/tmp/metadata_powerpack_YYYYMMDD.sql
```

2. Change the backup file location. Confirm that the backup file has been created at the chosen location.
3. Now go to `silo_mysql` and check the record in the DB (there will be one row present).

```
SELECT file_id, ppguid, name, revision FROM master_filestore.metadata_powerpack where file_id=FILE_ID and ppguid = 'PP_GUID';
```

4. Delete the record from the DB.

```
DELETE FROM master_filestore.metadata_powerpack where file_id=FILE_ID and ppguid = 'PP_GUID';
```

5. Confirm that the record has been deleted (there will be zero rows).

```
SELECT * FROM master_filestore.metadata_powerpack where file_id=FILE_ID and ppguid = 'PP_GUID';
```

6. Try to import the PowerPack. All the data on the installer should be updated according to the new version. Then install the PowerPack.

Refer to EM-52645: Xen PPv103 fails to install CLOSED for more information.

“Error: PowerPack Installation Failed”: Duplicate Primary Key

Possible Log Error: (em7php.log) |warning|error:Duplicate entry '477' for key 'PRIMARY' 07:46:10.25703|*pshare/classes/system/_powerpack/ppack_item.inc:error|771 |error|item insert failed

Check that the auto-increment setting for the `master.system_credentials` table is correct. First, find what the maximum `cred_id` is in `master.system_credentials`:

```
SELECT MAX(cred_id) FROM `system_credentials`
```

Then look to see what the next value for the auto increment field is:

```
SHOW CREATE TABLE master.system_credentials;
```

Look at the bottom of the output. If this number is less than the largest `cred_id` value, this is the problem. This can easily be fixed with the following SQL command:

```
ALTER TABLE master.system_credentials AUTO_INCREMENT = 1
```

Refer to EM-41972: PowerPack installation getting failed CLOSED for more information.

Remedy for Database lock when removing Powerpacks in version 11.3

This issue is related to the DB schema which is not owned by Sprint Eastwood.

Background

Due to a missing index on the master.dynamic_app_index_map.app_id column, 11.3 can cause database lockwaits (for example, see below: PowerPack Engine Troubleshooting Guide | 2024 09 25 EM 67099: erros messages and taskmanager seems to have some problem) during Powerpack removal if sufficiently high record counts exist in the master.dynamic_app_index_map table. Prior to 11.3, these records remained in the table after a Powerpack delete operation. The remedy is to assist the customer with the following .sql file to create the index (and optionally clean up old records that persist from failed/unexecuted delete statements).

File

daim_index_bat...ocs.sql
13 Sep 2023, 03:53 PM

File details

One .sql file, daim_index_batchdel_procs.sql, containing two MySQL stored procedure definitions:
p_create_daim_appid_index()
p_batchdelete_by_appid(APPID)

Command to install

`silo_mysql master < daim_index_batchdel_procs.sql`

Pre-execution steps

Prior to executing the created procedures, stop all services with “**sudo siloctl stop**” (for HA with failover, put the cluster in maintenance mode beforehand “**crm configure property maintenance-mode=true**”). If a deadlocked condition already exists, find which queries need to be manually killed with “**silo_mysql -e “show processlist” | grep Query | sort -rn -k 6 | head -20**”. Any delete queries for master.dynamic_app_index_map that get killed here should be added to a list so the app_id’s of the where statements can be used in the “p_batchdelete_by_appid” procedure call later.

Description

Procedure "p_create_daim_appid_index" will create a single-column index on master.dynamic_app_index_map.app_id if it doesn't already exist. This should be run while the database load is at a minimum. SE should recommend suspending collections while the index is being created. Example execution from CLI on the database: **silo_mysql master -e 'call p_create_daim_appid_index()'**. Index creation can take up to 15 minutes depending on table size.

Once the index is created, the customer should be able to delete powerpacks. Any leftover records in master.dynamic_app_index_map table from killed deadlock DELETE statements can be manually deleted with a simple delete statement (e.g. silo_mysql master -e 'delete from master.dynamic_app_index where app_id = 1001').

Some customers and SE's had to kill dynamic_app_index_map DELETE queries to break deadlocks. If the app_id's of these interrupted DELETE statements are known, the second stored procedure, "p_batchdelete_by_appid", is provided to remove those even before the app_id column index is added. Alternatively, this procedure may be helpful in the event of a deadlock while adding the index. The procedure can batch-delete master.dynamic_app_index_map records given the app_id parameter (e.g. silo_mysql master -e "call p_batchdelete_by_appid(1499)"). If adding the index takes too long, the intent of this procedure is to delete the unindexed dynamic_app_index_map records in smaller batches for app_id's that are no longer in use. SE will have to determine unused app_id's.

Examples

Call procedure to add index from CLI

```
$ silo_mysql master -e 'call p_create_daim_appid_index()'
+-----------------------------------------------------------------+
| status_msg                                                      |
+-----------------------------------------------------------------+
| Index idx_master_daim_appid created on master.dynamic_app_index_map |
+-----------------------------------------------------------------+
```

Call procedure to batch delete records from CLI

```
$ APP_ID=3340
$ silo_mysql master -e "call p_batchdelete_by_appid(${APP_ID})"
$ silo_mysql master -e "call p_batchdelete_by_appid(1499)"
```

Post-execution steps

Following deadlock mitigation and index creation, services may be restarted with "**sudo siloctl start**" and failover cluster may be placed in operating mode with "**crm configure property maintenance-mode=false**". Then the customer may use the UI to delete powerpacks as usual.

Associated Jira issues

SOL-24883: Database locks up when removing PowerPacks (Aurora MySQL 5.7.12) CLOSED

EM-56308: [EM-56308][EM-54685] Database locks up when removing PowerPacks CLOSED

EM-54685: Database locks up when removing PowerPacks CLOSED

Powerpack installation error -Unknown column 'curl_request_pwd' in field list

Description:

The sensitive Fields feature:

- try to control all password of the credentials. it include SOAP credential.
- the header(**curl_headers**) soap credential are just **key:value** info.
- type of item **curl_headers** is been confused with soap password fields(**curl_**). as a site effect of it the curl_header are marked with sensitive fields and `curl_request_pwd` is added on headers.
- it generate the export pp have wrong fields on soap headers
- it generates the install exception: `Unknown column 'curl_request_pwd' in 'field list'`

work around

- We need to remove the password fields on SOAP Credential before export in order for the export to work:
  - Embedded Password [%P]

- HTTP Auth Password
- Password

- **Fix**
  - **Fixed on**: 12.2.0
  - **Replicates version**: >= 11.1.0

Ticket related to this issue:

Key	Summary	Type	Link
SOL-24849	Powerpack that contains credentials that have additional headers specified fails to import	User stories	SOL-24849: Powerpack that contains credentials that have additional headers specified fails to import DONE
SOL-24537	Error with SOAP credential importing from 11.1 → 11.2	bug	SOL-24537: Error with SOAP credential importing from 11.1 → 11.2 CLOSED

At the moment of edit an AP2 dashboard the changes are not reflected in the build/export section of the PP properties.

When a Powerpack that contains an AP2 dashboard has been edited or modified this kind of changes are not reflected in the build/export section of the PP properties, so in order to make sure that this changes should be recognized by the build/export section of the PP properties to create a new build is needed to do the following:

- Make sure that all the required changes has been applied correctly.
- Then go to the “properties” section of the Powerpack.
- Make a click to the “save“ button.
- Then go to the Build/Export section of the Powerpack.
- The “Create a new build“ link should appears.

The above changes are needed in order to trigger a new build, the “Create a new build“ link should appears not by making a click to the “properties“ section of the Powerpack, the “Create a new build“ link should be triggered by making a click to another easily editable field.

Tickets related to this issue.

KEY	SUMMARY	TYPE	LINK
EM-54868	Prompt user to create a new build of PowerPack when editing AP2 dashboard	Task	EM-54868: [Telia Cygate Case 00299783] Prompt user to create new build of PowerPack when editing AP2 dashboard DONE

Diagnosis and Mitigation

Powerpack builds not appearing on UI

This is known to occur after applying this RPM to fix a zero day vulnerability: web-ui-12.1.2-484.el8.x86_64.rpm

This KB article describes the fix for this issue: https://support.sciencelogic.com/s/article/15527

Theory

table:CDB:master.definitions_ppack_actions

```
CREATE TABLE `definitions_ppack_actions` (
  `action_id` tinyint(3) unsigned NOT NULL AUTO_INCREMENT,
  `descr` varchar(64) NOT NULL,
  PRIMARY KEY (`action_id`),
  UNIQUE KEY `name` (`descr`)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
/*!40101 SET character_set_client = @saved_
```

table:CDB:master.definitions_ppack_items

```
CREATE TABLE `definitions_ppack_items` (
  `type_id` tinyint(3) unsigned NOT NULL AUTO_INCREMENT,
  `type_name` varchar(100) NOT NULL,
  `license_restricted` tinyint(4) NOT NULL,
  `type_label` varchar(100) NOT NULL,
  PRIMARY KEY (`type_id`),
  UNIQUE KEY `type_name` (`type_name`)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;

```

table:CDB:master.definitions_ppack_config_fields

```
CREATE TABLE `definitions_ppack_config_fields` (
  `content_tbl` varchar(50) NOT NULL,
  `content_field` varchar(50) NOT NULL,
  `preserve` tinyint(4) NOT NULL DEFAULT '1',
  PRIMARY KEY (`content_tbl`,`content_field`)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;

```

table:CDB:master_dynamic_app_index_map

```
CREATE TABLE `dynamic_app_index_map` (
  `did` int(10) unsigned NOT NULL,
  `app_id` int(10) unsigned NOT NULL,
  `group_id` int(10) unsigned NOT NULL,
  `instance` varchar(512) NOT NULL,
  `ind` int(11) unsigned NOT NULL,
  PRIMARY KEY (`did`,`app_id`,`group_id`,`ind`),
  UNIQUE KEY `did_app_group_instance` (`did`,`app_id`,`group_id`,`instance`)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;

```

Rows in this table are deleted in queries like this:

```
DELETE FROM master.dynamic_app_index_map WHERE app_id='2929'
```

by the delete code in the PP engine.

Enable Selective PowerPack Field Protection

This is a mechanism that allows customers to declare db fields that should not be altered by a powerpack upgrade.

This was observed to break down in EM-64118 (see post mortem articles below), which one customer worked around by using this query:

```
INSERT INTO
   `master`.`definitions_ppack_config_fields`
   (`content_tbl`, `content_field`, `preserve`)
VALUES
   ('master.dynamic_app_alerts;', 'state', '1');
```

However, even this work around failed to work when upgrading from SL1 from 11.3.0 to 12.1.1

Post Mortem Articles

2024-10-24 EM-70827 12.1.2 | Errors in RBA regarding api_support

LInk: EM-70827: 12.1.2 | Errors in RBA regarding api_support CLOSED

From the description:

```
Observed Behavior:
Some customer powerpacks were failing after upgrade to 12.1.2 - this was a while ago and prior to finding the defect.

Original case was 00461724 and slack thread https://sciencelogic.slack.com/archives/CQDFNG6EN/p1725999671206029
Result:{'debug_log': '
RBA started.
Event message: IBM i Device Found IBMI74.LOCAL | V7R4M0
Iseries version string: 7.4
Iseries hostname string: IBMI74.LOCAL
Find class SQL: SELECT class_type, class, descript FROM master.definitions_dev_classes WHERE class=\'IBM\' AND descript=\'IBM i 7.4\'
Found device class: (6722, \'IBM\', \'IBM i 7.4\')
Exception calling EM7 API ex: sequence item 0: expected string, NoneType found
Exception running RBA: Traceback (most recent call last):\n File "", line 200, in \n File "", line 125, in updateDeviceClass\n File "", line 116, in api_call\nAPIError: sequence item 0: expected string, NoneType found\n
RBA end.'}

Expected Behavior:
RBAs should succeed

Summary of troubleshooting steps taken so far and the current status:
In previous case, discussed this with engineering but since this was all custom content the customer/PS began trying workarounds we did not escalate. I was eventually alerted to the above defect / drop file and got in touch with the customer.
Currently waiting on details about what they've done to get this working, but Matt had me run a quick test and sees they do need the drop file. Here is the screenshot of the test.

```

2024-10-02 EM-67209: "Build / Export" pane not loading when you try and edit a PowerPack

Link: EM-67209: "Build / Export" pane not loading when you try and edit a PowerPack CLOSED

From the description:

```
Observed Behavior: After installing the web-ui-12.1.2-484.el8.x86_64.rpm, the PowerPacks build/export page is displaying no information under Manage PowerPack.
Expected Behavior: Ability to build and export PowerPacks after applying RPM.
Summary of troubleshooting steps taken so far and the current status:
Gathered screenshot, em7php.log, php-error.log, and mysqli.log debug while replicating the issue.
Similar to reported issue in https://sciencelogic.atlassian.net/browse/EM-67174
```

2024-10-01 EM-67174: Powerpack builds not appearing on UI post rpm install to patch the latest zero day vulnerability

link: EM-67174: Powerpack builds not appearing on UI post rpm install to patch the latest zero day vulnerability CLOSED

From the description:

```
Summary:

Datacom followed the KB article FAQ for Zero-Day Remote Code Execution

https://support.sciencelogic.com/s/article/15465 and applied the rpm on their 12.1.1 system hosted in AWS using DEs setup, customer Datacom 12.1.1 OL8 is noticing since installing the web-ui-12.1.2-484.el8.x86_64.rpm fix
powerpack builds have disappeared from the UI On inspecting the page, we observed - "Uncaught TypeError: $(...).html() is null" errors indicating something wrong with the UI.

Uncaught TypeError: Cannot read properties of null (reading 'split')
    at license_for (index.em7?sub=build&exec=powerpack_editor&ppid=390:339:42)
    at HTMLDocument.<anonymous> (index.em7?sub=build&exec=powerpack_editor&ppid=390:297:3)
    at Object.resolveWith (jquery.js?v=755:20:14257)
    at Function.ready (jquery.js?v=755:20:9242)
    at HTMLDocument.y (jquery.js?v=755:20:13725)

Devlog attached
em7php (1).log
01 Oct 2024, 03:54 AM
```

2024-09-25 EM-67099: erros messages and taskmanager seems to have some problem

ts_guide_to_jira_link

Link: 💔EM-67099: PowerPack deletion of DAs causes wait contention that causes event engine failures DONE

From the description:

```
Summary:
Slack:- https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1727262437777319
SL1 v11.3.2
Customer seeing error messages and event engine and taskmanager seems to have some problem.
Users cannot ack the events, not possible to create external tickets as well. Issue started today and this is cyclic now. Please suggest.
EM7 critical event: Query Failure in event engine ((1205, 'Lock wait timeout exceeded; try restarting transaction')): INSERT INTO master_events.events_active
Got lot of critical error messages in system_logs. Some RBA are not working and this is an error messages from system_status on the CDB:

    MySQL Queries running: 383 [KB 1275]
    Found multiples of the same query running more than 5 times: 63      UPDATE `config_push`.`table_status` SET `update_time` = NOW()
        WHERE `tbl_id` = 1566
    17      UPDATE master_events.events_active SET notify_count=notify_count+1, notify_stamp='2024-09-25 07:47:0
    9       UPDATE master_events.events_active
                    SET date_first='2024-09-11 09:06:06',
7       UPDATE master_events.events_active

SET date_first='2024-09-11 09:08:08',
[KB 1275]

    Currently there are 165 Sleep queries running [KB 1275]
```

This comment asked for an index to be added which solved the issue: 💔 EM-67099: PowerPack deletion of DAs causes wait contention that causes event engine failures DONE

```
Let’s get an index on the master.dynamic_app_index_map table on the customer’s CDB.  Request change approval with the customer for performing the following SQL statement on the DB:
ALTER TABLE master.dynamic_app_index_map ADD INDEX `app_id`(`app_id`)

Afterwards, run this to make sure the app_id index is present:
SHOW INDEXES FROM master.dynamic_app_index_map

Then they can delete the PowerPack, they were attempting to delete, if they still need to do so.
@Chris Marshall Let me know if this interferes with any of your troubleshooting.
```

This comment identified the problem as originating with the powerpack engine: 💔 EM-67099: PowerPack deletion of DAs causes wait contention that causes event engine failures DONE

Here is a quote of the summary section of that comment:

```
Summary
    support gathered seven hours of watch_locks data
    which contained a five minute (300 second) stretch of time where the problem of long held lockwaits occurred
    the central query is the same one we saw before:
       DELETE FROM master.dynamic_app_index_map WHERE app_id='2929'

we now know this is originating from power pack engine code responsible for deleting power pack data:
https://github.sciencelogic.com/PowerPacks/pp_build_engine/blob/115851c013d4404da75d4f3601a902dfa7b8091a/G3code/phpshare/classes/system/_powerpack/_powerpack_app.inc
The customer remarked (which also points to the PP engine code)
   It seems it's happening when trying to delete a now unused powerpack (Dell EMC Vmax). We had a lot of those devices and had problem
    when doing the recursive delete. We had to delete them not from the component devices but directly from the registry.

    I expect that the PP engine code for deleting PP data is
        starting a multi query transaction
        deleting any active events that match a policy id owned by the powerpack
        deleting rows of dynamic_app_index_map for app_ids owned by the powerpack
        accumulating locks across those queries
        which can then interfere with lots of other queries in the system (like the event engine, which has crashed as a result of this)

    we have two things to look into here
        why the delete queries against master.dynamic_app_index_map are taking so long (would adding an index on app_id help?  there is not one presently)
        does the powerpack engine need to use multi-query transaction when deleting a powerpack?  they are a recipe for lock waits when one of the queries gets stuck

    I am going to ask sprint eastwood (the owners of the PP engine code) to take over from here
        If they could, during their analysis, point out the mechanism in PHP code whereby multiquery transactions are enabled
        I would be much obliged as I don’t know how that’s done and I suspect multiquery transactions in php code are behind a number of other recruing issues
```

2024-09-05 EM-66709: Cisco Onboarding PP Failes on EM7 12.1.2 version.

ts_guide_to_jira_link

Link: EM-66709: Cisco Onboarding PP Fails on EM7 12.1.2 version. CLOSED

From the description:

```
Observed Behavior: Cisco has at least 4 PP where execution environments are not being built due to could not build EE errors. I know we have current issues with 12.1.2 and CDW and we feel this is the same type
of issues but these apps are not using silo_apps.

Expected Behavior: PP EE should be able to be built and allow the DA's to run.

Summary of troubleshooting steps taken so far and the current status:
When running the EE in deugg we get the error:
Exception: b"WARNING: The directory '/opt/em7/.cache/pip' or its parent directory is not owned or is not writable by the current user. The cache has been disabled. Check the permissions and owner of that directory.
If executing pip with sudo, you may want sudo's -H flag.\nLooking in links: /opt/em7/envs/uenv-3177890073031647365/pypi\nProcessing /opt/em7/envs/uenv-3177890073031647365/pypi/xlrd-2.0.1-py2.py3-none-any.whl\nERROR:
Could not find a version that satisfies the requirement cisco_onboarding==9.3.0.1 (from -r /opt/em7/envs/uenv-3177890073031647365/requirements.txt (line 6)) (from versions: none)\nERROR: No matching distribution found
for cisco_onboarding==9.3.0.1 (from -r /opt/em7/envs/uenv-3177890073031647365/requirements.txt (line 6))\n"
This seems to be due to the library not being in a wheel file and in a PY_directory built library. Our current documentation does not clearly state how these need to be built for 12.1.2.
If we could get the steps to correctly build these libraries in an KB then we can supply the KB to the customer telling them to fix each PP they encounter this on.
```

2024-08-01 EM-66232: Not able to import v105 of the Restorepoint PP

ts_guide_to_jira_link

Link: EM-66232: Not able to import v105 of the Restorepoint PP CLOSED

From the description:

```
Observed Behavior:
Not able to import v105 of the Restorepoint PP

Expected Behavior:
Should be able to import v105 of the Restorepoint PP

Summary of troubleshooting steps taken so far and the current status:
the import of the Restore Point v105 PowerPack was failing with this  "ERROR: Could not unpack PowerPack file for installation", when I checked the em7php.log file I spotted this could not unpack source with provided
encryption key error, I've found duplicate entries for version 105 with the same revision in the master_filestore.metadata_powerpack table, I found a reference to that error
on this https://sciencelogic.atlassian.net/wiki/spaces/ENG/pages/27999698966/PowerPack+Engine+Troubleshooting+Guide TSG but I believe we made a mistake we deleted both duplicate entries when it should be only one I
think, we didn't make a backup of the master_filestore.metadata_powerpack table as well, now when we trying to import v105 of the Restore Point PowerPack it doesn't present any error, it just closes the Import PowerPack
pop up and then when we go to Actions > Install PowerPack on the PowerPacks page v105 is not in the list of the imported PowerPacks, I collected the mysqli and em7php.log files with debug but I can't see any error
there for the import issue, I just see an could not unpack source with provided encryption key error entry but that's for the initial issue, I can see two duplicate entries in the  master_filestore.metadata_powerpack
table for v104 and one has the same revision as version 105 i.e 72, could it be this entry that is preventing the import? and on the master.powerpack table I only see one entry for the restorepoint PP v104.

Discussed this issue with eng on this https://sciencelogic.slack.com/archives/C077D8K3T3Q/p1722495618428089 slack thread and an escalation was requested.
```

The slack thread is here: https://app.slack.com/client/E03R7NUFC4X/C077D8K3T3Q

2024-06-28 EM-65627: License is not valid when use Administrative license key type on powerpack installation

ts_guide_to_jira_link

Link: EM-65627: License is not valid when use Administrative license key type on powerpack installation CLOSED

From the description:

```
Observed Behavior:
Scenario 1:
Build Install-only PowerPack from OOTB PP - generate Admin Key 88-char length
Import Build - apply Admin Key (License Key is Invalid)
Scenario 2 - Following On From Last Scenario
Re-Build now with Admin Embedded License - Generate Admin Key 64-char
Import Build - Apply Admin Key -> Successful
Scenario 3 - Follow On Again From Last Scenario (Same Steps as First)
Re-Build PP Again with Install-Only Embedded License - Generate Admin Key 64-char
Import Build - Apply Admin Key -> Successful

A dropfile was provided already in this Escalation, where they said the issue was resolved: https://sciencelogic.atlassian.net/browse/EM-58215
However, this was only resolved in Scenario 3 (which is the same as 1) but this means they always have to build the PP first as Embedded License as Admin
```

2024-06-10 EM-65354: Build and Delete of customized PowerPack fails

ts_guide_to_jira_link

Link: EM-65354: Build and Delete of customized PowerPack fails CLOSED

From the description:

```
Observed Behavior:
Customer has customized a PowerPack and is unable to compile to create a new build or delete the PowerPack. The customization has been performed on the Microsoft SQL Server Enhanced version 108. When attempting to
compile or delete the debug logs show the error "Failed to initialize item of type [app] w/ identifier [1860]". We have been unable to identify an app, alert, or event in the PowerPack with the ID connecting to 1860.

Expected Behavior:
To be able to compile or delete the PowerPack.

Summary of troubleshooting steps taken so far and the current status:

Verified on the platform that Dynamic Application with the ID of 1860 does not exist. Reviewed Event Polices and alerts to see if any were configured with a Dynamic Application that does not exist or one associated
with 1860 and were unable to locate any. Customer was able to identify an app_id of 1860 located in the dynamic_app_index_map and so far this is the only 1860 identified in the database.
```

2024-05-15 EM-64989: Not able to import SQL server powerpacks

ts_guide_to_jira_link

Link: SOL-27339: Not able to import SQL server powerpacks CLOSED

From the description:

```
Observed Behavior:Not able to import few powerpacks

Expected Behavior:customer should be able to import and install powerpack

Summary of troubleshooting steps taken so far and the current status:

1)Went on a webex with customer and cleared SL1 cache and asked customer to upload PP.I observed the moment customer was trying to import powerpack he got error while uploading powerpack
2)Enabled PHPdevlog and observed unknown error in mysqli.log and below error in em7php.lgo

13:11:35.08852|/opt/em7/gui/ap/classes/dashboard_widget.inc:obj      |131 |debug|                    [UNKNOWN] => "Unknown error"
13:11:35.08854|/opt/em7/gui/ap/classes/dashboard_widget.inc:obj      |131 |debug|                    [EM7_ERROR] => "Unknown error"
13:11:35.08856|/opt/em7/gui/ap/classes/dashboard_widget.inc:obj      |131 |debug|                    [PPK_NOT_EDITABLE] => "Current PowerPack license does not allow alteration of this element"
13:11:35.08858|/opt/em7/gui/ap/classes/dashboard_widget.inc:obj      |131 |debug|                    [ENT_INVALID_FIELD] => ""

3) uploaded the same powerpack in our lab and generated with admin key and shared with customer , but it got failed again.

4) attaching all the logs
```

2024-05-10 EM-64908: Building powerpack includes lot of unwanted content libraries

ts_guide_to_jira_link

Link: EM-64908: Building powerpack includes lot of unwanted content libraries CLOSED

From the description:

```
Observed Behavior: Complied PP have apps, alerts, events and execution envionments that should not be included. The customer is looking for a process to edit the PP outside the UI and recompile so the unwanted items are not installed onto the syste,

Expected Behavior: A process to edit and recompile the PP outside the UI.

Summary of troubleshooting steps taken so far and the current status:

When installing the PP even though the install says ignore on the unwanted items they are still installed. There must be some ref to thes eitems in the wanted DA's but we do not have a documented process that the customer to use to remove the unwanted itmes and recompile.

When the PP is installed thes unwanted items are not linked to the PP so you can't just remove them from the PP and rebuild using the UI.
```

2024-05-01 EM-64746: Need drop file for 12.1.1 to resolve PPK deletion issue in EM-62958

ts_guide_to_jira_link

Link: EM-64746: Need drop file for 12.1.1 to resolve PPK deletion issue in EM-62958 CLOSED

From the description:

```
Reason for Escalation:
Engineering assistance needed to diagnose issue in order to find workaround
(requesting drop file approval for 12.1.1 SaaS stack)

Observed Behavior:
Deleting the VMware PPK caused an outage due to 1000's of component device deletes and DB deadlocks

Expected Behavior:
Deleting a PPK should work better, and drop file seems to fix this

Summary of troubleshooting steps taken so far and the current status:
Slack channel: #outage-alsac-00433686
Leo posted a lot of screenshots and outputs to above channel.

We were able to restore stability after restarting php-fpm a couple times and telling customer to stop deletions.
Recommended to delete in batches of 50-100 at most.

Now, we just need the drop file approved for the record and get SRE to load that onto the DEs.
```

This is a request for the same drop file developed in EM-62958 (for an 11.3.0 system) to be developed for this customer (on a 12.1.1 system).

2024-04-01 EM-64118: Custom Fields for the Enable Selective PowerPack Field Protection Feature Not Being Preserved

ts_guide_to_jira_link

Link: 💔EM-67082: Custom Fields for the Enable Selective PowerPack Field Protection Feature Not Being Preserved CLOSED

From the description:

```
Observed Behavior: Custom fields added to the Enable Selective PowerPack Field Protection feature
are not preserving after PowerPack Upgrade as expected.

Expected Behavior: All fields in the Enable Selective PowerPack Field Protection database table
should be preserved.

Summary of troubleshooting steps taken so far and the current status:

CDWS is reporting that after upgrading the Cisco: Meraki [API] PowerPack, the alert state for
dynamic app alerts are not being preserved. Knowing that this is not a field that is protected
when the Enable Selective PowerPack Field Protection feature is enabled in the
System > Settings > Behavior page, the customer is running the SQL query
INSERT INTO `master`.`definitions_ppack_config_fields` (`content_tbl`, `content_field`, `preserve`)
VALUES ('master.dynamic_app_alerts;', 'state', '1'); to add the fields.

According to the customer, this is a practice they have been doing for years and it has been
working until they upgraded SL1 from 11.3.0 to 12.1.1. As a test, I reproduced this issue by
running the above SQL query in my lab system, enabling the Enable Selective PowerPack Field
Protection feature and upgrading the PowerPack and indeed the dynamic app alert state is not
being preserved. During our testing, we found the default fields and other custom fields are
being preserved so we are unsure why this specific field is not working. I did find an old
Jira from CDW for the same issue here https://sciencelogic.atlassian.net/browse/EM-16662
but the fix of clearing the SL1 system cache does not fix the issue this time. Requesting
assistance from engineering to determine what has changed and to figure out a way to restore
this functionality.

I attached a spreadsheet showing the output of
SELECT * FROM `master`.`definitions_ppack_config_fields`.
```

2024-01-19 EM-62958 Database locks up when removing PowerPacks due to master.policies_automation_events query

Link: 💔EM-62958: Database locks up when removing PowerPacks due to master.policies_automation_events query CLOSED

ts_guide_to_jira_link

From the description, this query seems to be causing the issue:

```
events_delete()/opt/em7/gui/ap/functions/functions_events.inc [line:16 calling:autofetch_column]

SELECT pae.ap_id, count(pae.eid) as num_events
FROM master.policies_automation_events pae
RIGHT JOIN master.policies_automation_events pae2 ON (pae.ap_id=pae2.ap_id AND pae2.eid='3848')
GROUP BY ap_id
HAVING num_events = 1

SIZE: 270 bytes
ROWS:0|ERRNO:0|MSG:
DURATION: 98.63682 seconds
```

Also in the description, this case seems to be related: SOL-24883: Database locks up when removing PowerPacks (Aurora MySQL 5.7.12) CLOSED

In this comment EM-62958: Database locks up when removing PowerPacks due to master.policies_automation_events query CLOSED

the instructions for deploying the drop file

```
functions_events.inc
```

are laid out. This was developed for an 11.3.0 system.

2023-02-01 EM-56308: Database locks up when removing PowerPacks

Link: EM-56308: [EM-56308][EM-54685] Database locks up when removing PowerPacks CLOSED

ts_guide_to_jira_link

One of the comments in this case explains the theory:

```
The “bug” manifests when deleting objects with a lot of associated data. The threshold at which
DB lockups occur is dependent on the amount of data for a collection object, DB traffic, and
system resources. It would affect any system prior to the fix, regardless whether it’s
deployed on-prem or SaaS (with RemoteDB/RDS).

I put bug in quotes because for small enough datasets, the functionality is perfectly valid.
Prior to our fix, when deleting via a UI action the PHP executes a synchronous SQL DELETE
statement for all data associated to the collection object. The bug fix transfers this s
tatement to an async process to avoid locking up the UI or causing excessive UI latency,
which could happen if the statement succeeds but takes a while. The async process executes
multiple DELETEs to limit the number of records affected (and therefore limit the scope of
DB locks required) in each statement.

```

This issue may be related to EM-42726: Deletion of Collection Objects from Dynamic Application causes large quantity of delete queries t o lock up the database CLOSED

KB Articles

[Minor] DRAFT ARCHIVED

2023-01-24 EM-53608: Unable to delete Powerpack via API in 11.x Was able to delete in 10.x

ts_guide_to_jira_link

Link: EM-56086: Unable to delete Powerpack via API in 11.x. Was able to delete in 10.x CLOSED

From the description

```
Observed Behavior: With 10.2.x release you can do DELETE via API to delete a PowerPack. That
functionality does not work in 11.x anymore. Tested on 11.2.x and 11.3.0.

Expected Behavior: Need to be able to delete PowerPacks via API to be able to
```